



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 65: Solar Playground with Security System Bi-Weekly Update 3

**Team Members: Zayan Prasla, Alex Allahar, Jackson
Kitchel, Victor Igbokwe**
Sponsor: Wonhyeok Jang
TA: Zian Wang

Project Summary

The SolarEye Security System is a solar-powered, autonomous surveillance solution for playgrounds and parks.

It enhances security with motion-activated cameras, real-time mobile alerts, LED deterrent lighting, and visitor charging stations.

This eco-friendly system ensures continuous, cost-effective protection for private parks and properties.



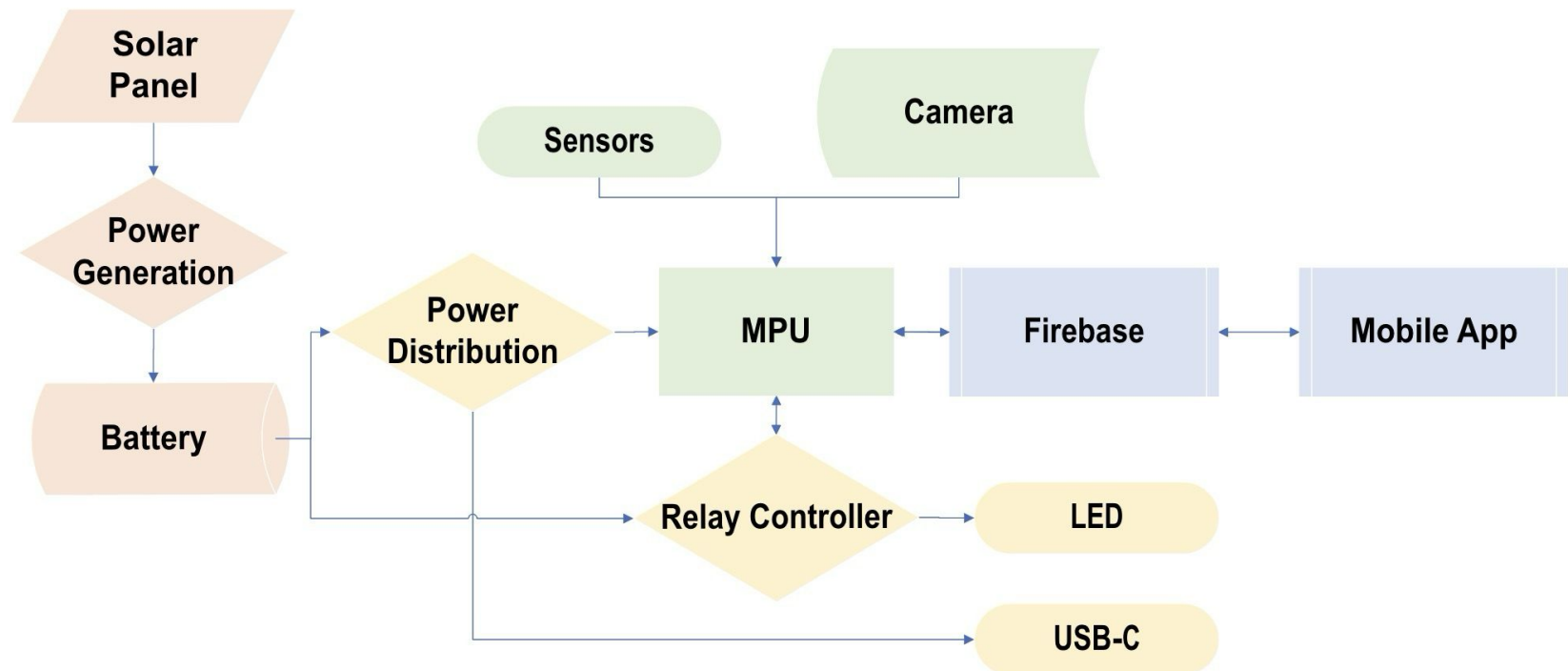
Project/Subsystem Overview

Red - Jackson Kitchel (Power Generation)

Yellow - Victor Igbokwe (Power Distribution)

Green - Alex Allahar (Microprocessor & Sensors)

Blue - Zayan Prasla (Network & Application)





Project Timeline

New PCBs Ordered/ Received (completed 02/03)	Integration of MPU and Application subsystems (to completed 02/25)	Integration of Power Delivery system with MPU (to complete by 02/25)	Integration of Power Generation System with Power Delivery System (to complete by 02/28)	Final Integration (to complete by 03/03)	System Test (to complete by 03/10)	Validation (to complete by 03/20)	Demo and Report (to complete by 04/25)
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Power Generation

Accomplishments since last update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Both boards soldered experienced issue of a capacitor burning up • Issue was found to be component rated for wrong amperage • New component arrives tomorrow	• Integrate with power distribution subsystem • Test Current and Voltage sensor outputs with MCU subsystem

Power Generation



Figure 1: Component Needing Replacement

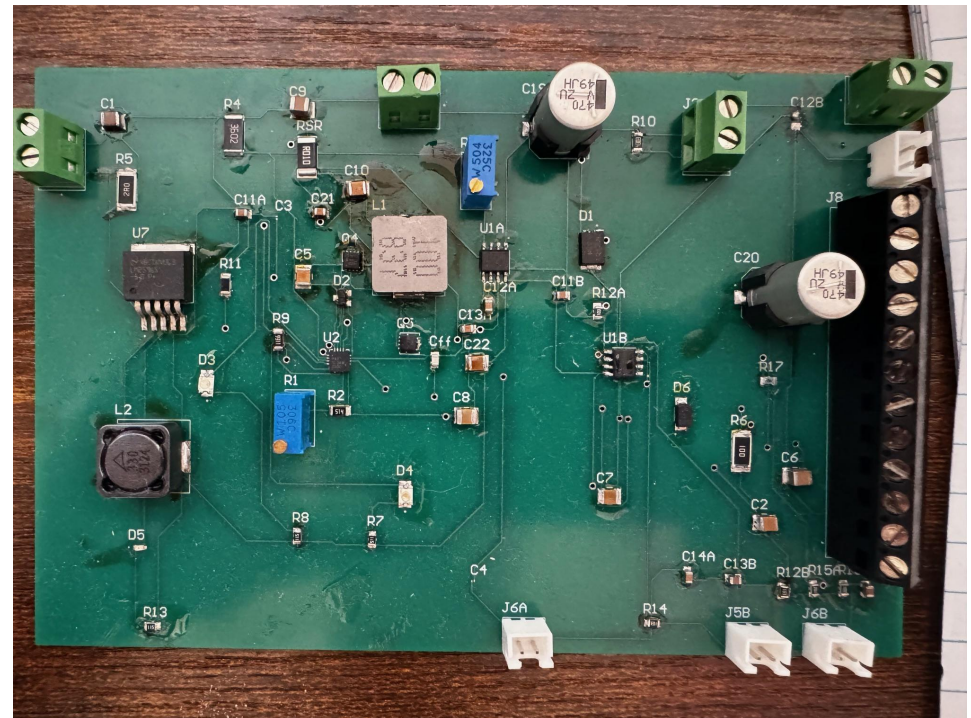


Figure 2: Power Generation PCB



Power Distribution

Accomplishments since last update 16 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Successful Integration with Raspberry Pi MPU ❖ Soldered Relay Control module PCB	❖ Validate Buck converter Integration for all modes of operation on MPU. ❖ Integration of relay control module PCB with Microprocessor system. ❖ Integration with

Power Distribution

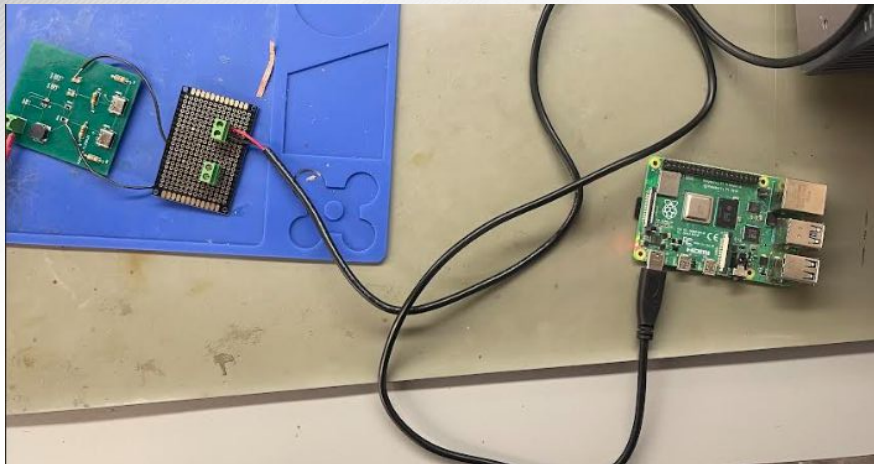


Figure 3: Buck converter to MPU Integration

Line Regulation test using Power Supply		
Output Current (controlled current, A)	Input Voltage (V)	Output Voltage (V)
2	11.4	5.08
2	11.8	5.07
2	12.299	5.07
2	12.998	5.06
2	11.4	5.08
2	11	5.08
2	10.801	5.08
2	10.6	5.08
2	10.001	5.09

Figure 4: Line & Load regulation Tables

5V Load regulation test w/PS	
Output Current (A)	Output Voltage (V)
0.021	5.089
0.051	5.088
0.102	5.085
0.151	5.084
0.25	5.07
0.401	5.07
0.701	5.068
1.001	5.065
1.201	5.05
1.4	5.03
1.601	5.02
1.65	5.02
1.85	5.004
2.01	4.989
2.09	4.974

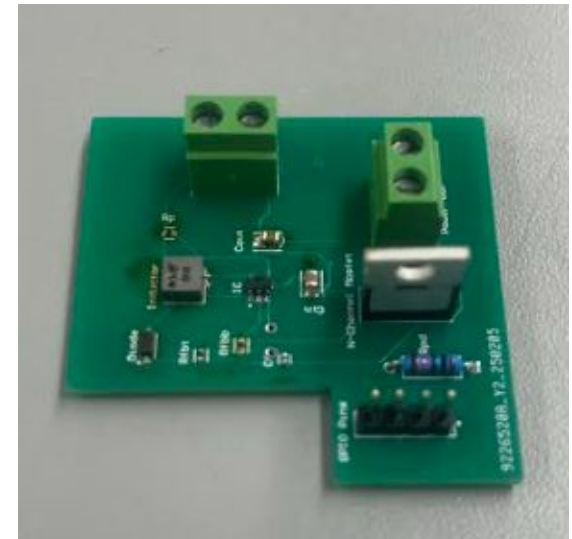


Figure 5: Soldered Relay Control module board



Microprocessor & Sensors

Alex Allahar

Accomplishments since last update 17 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Completed Integration of MPU with SolarEye App ❖ Integrated MPU with Buck Converter PCB ❖ Validated event detection to app delay	❖ Validate All Modes of Operation on Buck Converter PCB ❖ Validate Video Footage on Mobile App ❖ Complete Integration of Relay Controller PCB with MPU. ❖ Functionality Testing of App controlling Camera, LED, and sensor

Microprocessor & Sensors

Alex Allahar

Event Detected by MPU	Delay (s)
1	0.4922
2	0.1038
3	0.1146
4	0.0838
5	0.909
6	0.0893
7	0.0881
8	0.0981
9	0.0801
10	0.0793
MPU to APP Delay Avg:	0.21383

Figure 6: MPU to App Delay Table

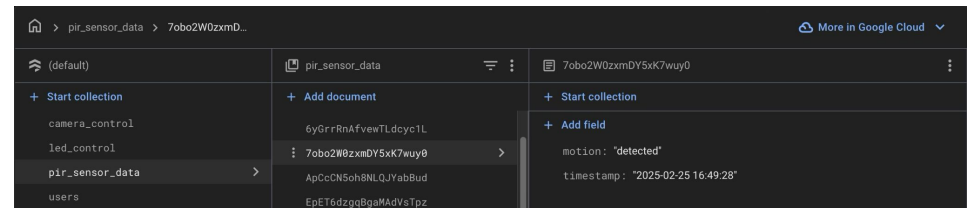


Figure 7: Firestore App Event History Database

```
pi@raspberrypi:~/solareye $ python sensor_data_firestore.py
Monitoring GPIO pin for sensor activation...
Motion detected at: 2025-02-25 16:49:28
Motion detected and data pushed to Firestore
pi@raspberrypi:~/solareye $
```

Figure 8: Event Detection Testcase Terminal

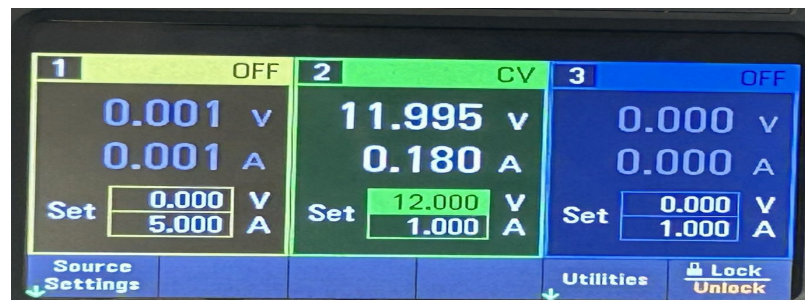


Figure 9: MPU Peak Current on Buck Converter PCB



Application & Network

Accomplishments since last update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Completed Integration between App and MPU ❖ Complete Camera/LED controls from app to Firebase to MPU ❖ Completed testing all backend security ❖ Instant updates to Event History	❖ Video and picture transfer from MPU to App ❖ Power level data communication (Battery Level Graph) ❖ Lower power mode ❖ Live Streaming ❖ Ongoing updates to the Frontend

Application & Network

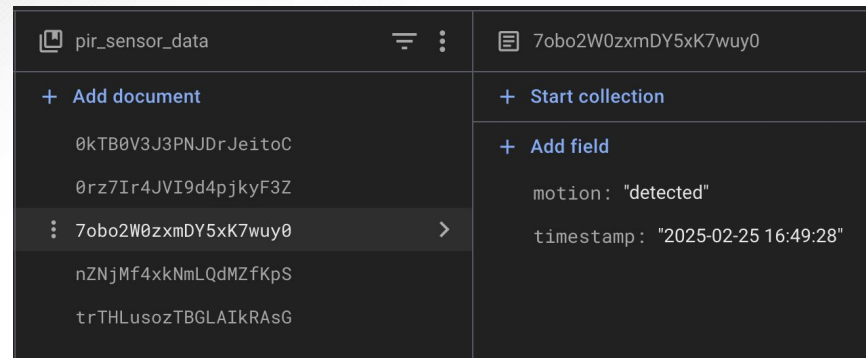
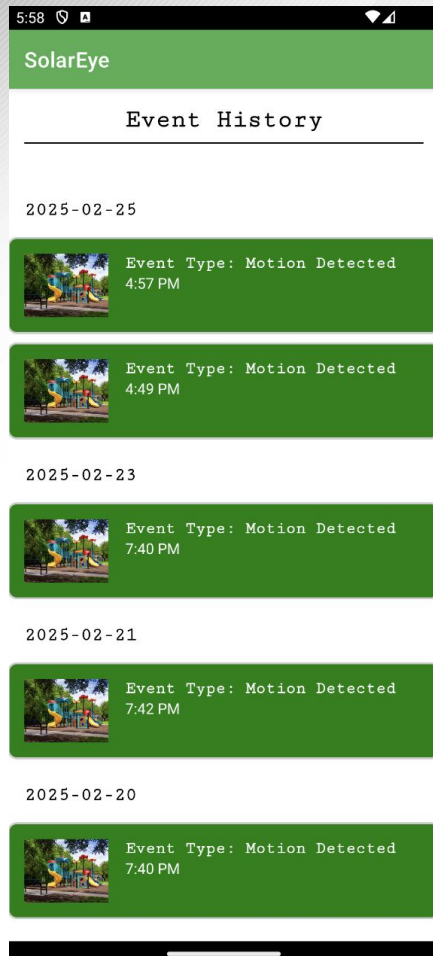


Figure 11: Motion Detection Database

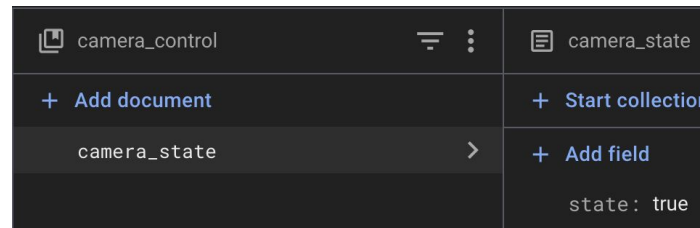


Figure 12: Camera Control Database

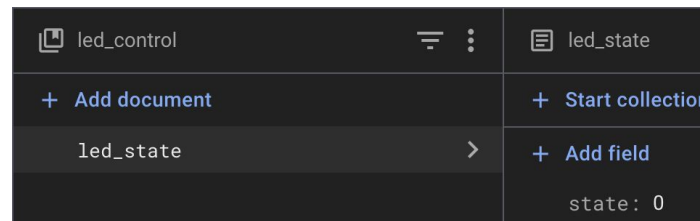


Figure 13: LED Control Database

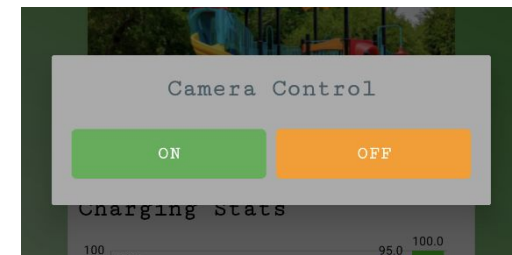


Figure 14: Camera Control on App

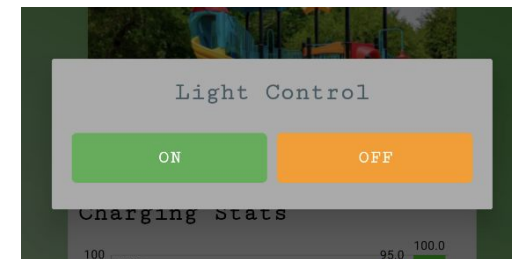


Figure 15: LED Control on App

Figure 10: Event History Page

Execution & Validation Plan

[illegible]



Execution & Validation Plan

Test Name	Success Criteria	Methodology	Status	Assigned to
LED Relay Controller MPU Control	Be able to control the LED from on to off state, or automatically if motion is detected from the MPU.	Use camera to send a 30-second video to Firebase, and determine time between video capture and firebase acknowledgement.	Partial Completion	Alex Allahar
Modes of Operation App Control	Be able to control all three modes of operation via the app	Using firestore database and GCP functions to send a update to the MPU changing the mode of operation.	Pending	Alex Allahar
App Event History	Both MPU and App has up to a week of footage.	Verify that the MPU and Firebase after reboot has the up to 24 of prevent footage.	Partial Completion	Alex Allahar
Video Streaming	The MPU live-stream can be access via App.	Use UDP to Stream the MPU via HTTP Flask Server.	Partial Completion	Alex Allahar
MPU to App Automatic Delay	Delay is below 1 s. To make it almost automatic.	Using PIR_sensor_data_firestore.py perform 10 testcases and average the delay.	Completed	Alex Allahar
Validate Buck integration with MPU	All modes of operation on MPU function while integrated wit buck converter PCB	Connect PCB and test all modes of operation	Partial Completion	Victor Igbokwe
Buck LineTest	Output voltage remains constant with different input voltages	Change input voltage and record output voltages	Completed	Victor Igbokwe
Buck Load Test	Output voltage remains constant as the output current increases.	Use Electronic load device	Completed	Victor Igbokwe
Relay control module Integration	Current flow can be controlled by MPU	Connect GPIO pins on PCB to MPU	Pending	Victor Igbokwe
Buck Converter integration with Solar Generation subsystem	Buck converter powered by solar generation system	Connect buck PCB to battery	Pending	Victor Igbokwe
Network Connectivity	Able to receive and send signals to and from the app/system.	MPU data is able to be sent and seen on the app.	Completed	Zayan Prasla
Password Policy	User has to put in a password that follows the password policy	Check if password can be created by not following the policy: 6 minimum characters, 1 uppercase and 1 special character	Completed	Zayan Prasla
Store and Fetch Data	Application is able to store and fetch login data.	Test logging into the app with an already created account	Completed	Zayan Prasla
Email Verification	After creating an account, user recieves a verification email.	Check for a verification email being sent after creating an account	Completed	Zayan Prasla
Main Page UI	Successfully display Event History, Charging statistics, and system controls	Check each function on main page and make sure there is no input lag	Completed	Zayan Prasla
Solar Generation Check	A charge rate of 0.45kW.	Test solar charge rate with a multimeter.	Pending	Jackson Kitchel
Operation Time	8 hours of battery life	Monitor the battery after it finishes fulling charging.	Pending	Jackson Kitchel
High Voltage Cut-Off	Charging circuit cuts off when voltage level reaches 15V	Monitor the charging voltage after battery is fully charged	Pending	Jackson Kitchel
Low Voltage Cut-Off	Battery starts charging when low-voltage of 12.4V	Monitor the charging voltage after charge level is low	Pending	Jackson Kitchel
PCB Functionality	PCB Soldered and functioning as required by previous checks	Validate every requirement after PCB is complete	Partial Completion	Jackson Kitchel
PCB Design	PCB Design is plausible.	Simulation Results using software	Completed	Jackson Kitchel



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Thank You